



Lab 09: WLAN vs Wireless

WRT300N Wireless Router · Router0 (2911) · Switch0 · Printer · 3 PCs · Laptop

Lab #09

Topic: Wireless Networking &
WLAN

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build a combined wired LAN and wireless WLAN in a single Packet Tracer topology.
- Configure a Linksys WRT300N wireless router with SSID 'Yasin' and WPA2 security.
- Connect Laptop0 wirelessly to the WRT300N using PSK 'ahmed1234'.
- Connect PC0, PC1, PC2 and Printer0 to Switch0 on the wired LAN.
- Connect Router0 (2911) to WRT300N to provide inter-network routing.

2. Network Topology

Switch0 (2960-24TT) is the wired backbone with Printer0 (port 0), PC0 (Fa0/1), PC1 (Fa0/2) and PC2 (Fa0/3). Switch0 connects to Router0 (2911) on Fa0/4 → Gig0/0. Router0 connects to WRT300N Wireless Router on Gig0/1 → 0/0 (dashed = wireless). Laptop0 connects wirelessly to the WRT300N using SSID 'Yasin'.

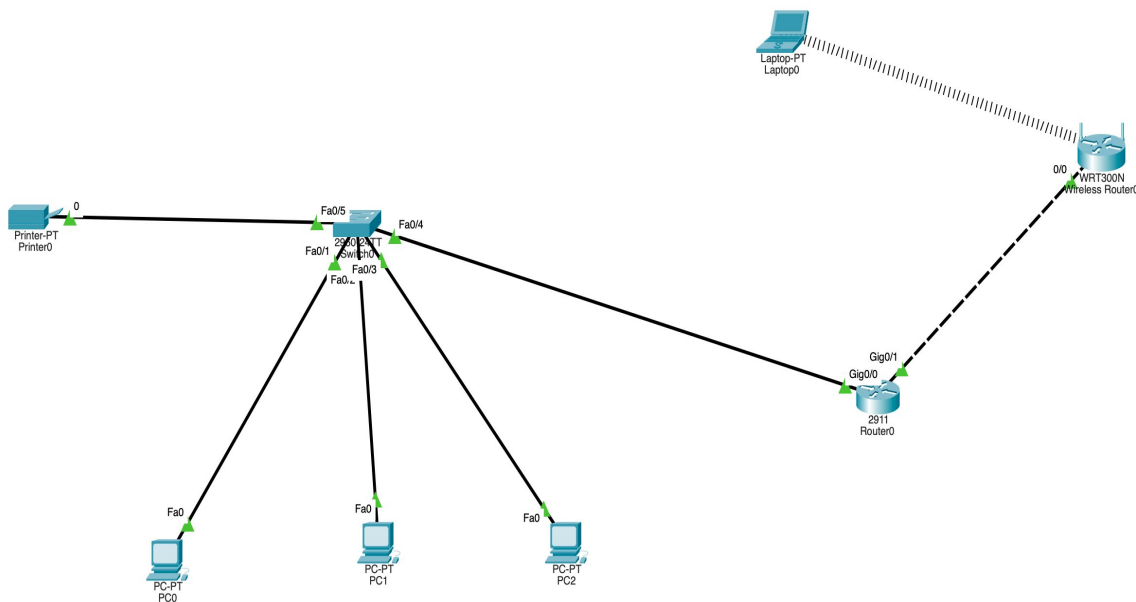


Figure 1 – Full topology: wired LAN (Switch0, PCs, Printer) + wireless (WRT300N, Laptop0)

3. WRT300N Wireless Router Configuration



3.1 Network Setup (Basic Setup tab)

Access WRT300N → GUI → Setup → Basic Setup:

Figure 2 – WRT300N Basic Setup: Router IP 192.168.2.1, DHCP pool 192.168.2.100–149

Router IP	192.168.2.1
Subnet Mask	255.255.255.0
DHCP Server	Enabled
Start IP	192.168.2.100
Max Users	50
IP Range	192.168.2.100 – 192.168.2.149

3.2 Wireless Settings



Wireless-N Broadband Router

Firmware Version: v0.93.3

Setup | Setup | **Wireless** | Security | Access Restrictions | Applications & Gaming | Administration | Status

Basic Setup | DDNS | MAC Address Clone | Advanced Routing

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers):

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: 192 . 168 . 2 . 1

Subnet Mask: 255.255.255.0

DHCP Server Settings

DHCP Server: ☒ Enabled ☐ Disabled

DHCP Reservation

Start IP Address: 192.168.2. 100

Maximum number of Users: 50

IP Address Range: 192.168.2. 100 - 149

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0

Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

WINS: 0 . 0 . 0 . 0

Help...

Figure 3 – WRT300N Wireless tab: SSID 'Yasin', Channel 1 (2.412 GHz), Mixed mode, SSID broadcast ON

SSID	Yasin
Network Mode	Mixed
Radio Band	Auto
Standard Channel	1 – 2.412 GHz
SSID Broadcast	Enabled

3.3 Wireless Security



Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless Setup Wireless Security Access Restrictions Applications & Gaming Administration Status

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter Advanced Wireless Settings

Basic Wireless Settings

Network Mode: Mixed

Network Name (SSID): Yasin

Radio Band: Auto

Wide Channel: Auto

Standard Channel: 1 - 2.412GHz

SSID Broadcast: ☒ Enabled ☐ Disabled

Help...

Figure 4 – WRT300N Wireless Security: WPA2 Personal, AES, Passphrase ahmed1234

Security Mode	WPA2 Personal
Encryption	AES
Passphrase	ahmed1234
Key Renewal	3600 seconds

3.4 Wireless Interface Settings (from Packet Tracer config)

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless Setup Wireless Security Access Restrictions Applications & Gaming Administration Status

Basic Wireless Settings Wireless Security Guest Network Wireless MAC Filter Advanced Wireless Settings

Wireless Security

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: ahmed1234

Key Renewal: 3600 seconds

Help...



Figure 5 – WRT300N wireless interface: SSID Yasin, WPA2-PSK, AES, PSK ahmed1234

4. Router0 (2911) Configuration

```
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface GigabitEthernet0/1
Router0(config-if)# ip address 192.168.2.254 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
! Static route so wired LAN can reach wireless subnet
Router0(config)# ip route 192.168.2.0 255.255.255.0 192.168.2.254
Router0# write memory
```

5. Verification

```
Laptop0> ping 192.168.2.1          ! Laptop to WRT300N gateway
Laptop0> ping 192.168.1.10        ! Wireless to wired (via Router0)
PC0> ping 192.168.2.100           ! Wired PC to wireless Laptop
Router0# show ip route             ! Verify both subnets present
```

6. Key Takeaways

- The WRT300N is a home-grade wireless router with built-in DHCP — different from an enterprise AP.
- SSID 'Yasin' is broadcast so clients can discover the network automatically.
- WPA2 Personal (PSK) with AES is the correct security setting for a small wireless network.
- A wired LAN and WLAN can coexist on different subnets and communicate via a router.
- The printer on the wired LAN can be accessed from wireless clients only when routing is configured.